

Step 1: Create Project Directory

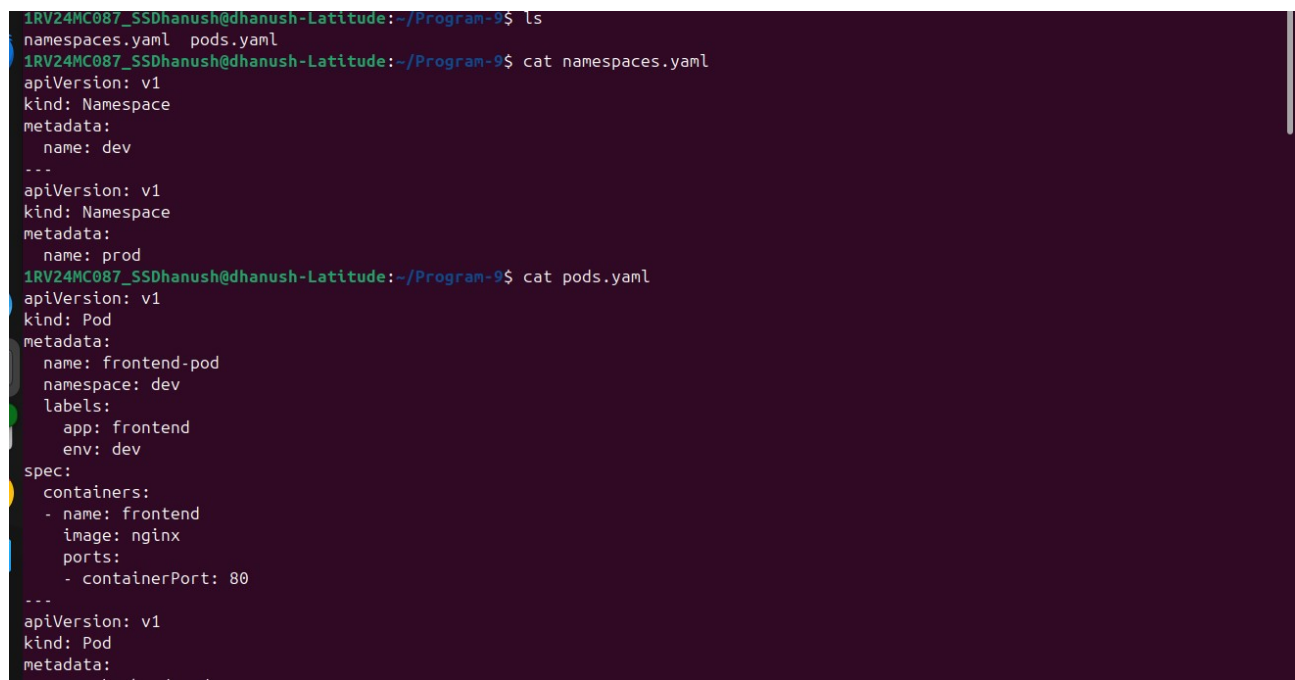
```
mkdir Program-9  
cd Program-9
```

Step 2: Create namespaces.yaml

```
nano namespaces.yaml
```

Step 3: Create pods.yaml

```
nano pods.yaml
```

A terminal window with a dark purple background. The prompt is 1RV24MC087_SSDhanush@ghanush-Latitude:~/Program-9\$. The user runs 'ls' and sees 'namespaces.yaml' and 'pods.yaml'. Then they run 'cat namespaces.yaml' and see a YAML file with two namespace definitions: 'dev' and 'prod'. Next, they run 'cat pods.yaml' and see a YAML file with two pod definitions: 'frontend-pod' in the 'dev' namespace and an unnamed pod in the 'prod' namespace. The 'frontend-pod' uses the 'nginx' image and port 80.

```
1RV24MC087_SSDhanush@ghanush-Latitude:~/Program-9$ ls  
namespaces.yaml  pods.yaml  
1RV24MC087_SSDhanush@ghanush-Latitude:~/Program-9$ cat namespaces.yaml  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod  
1RV24MC087_SSDhanush@ghanush-Latitude:~/Program-9$ cat pods.yaml  
apiVersion: v1  
kind: Pod  
metadata:  
  name: frontend-pod  
  namespace: dev  
  labels:  
    app: frontend  
    env: dev  
spec:  
  containers:  
    - name: frontend  
      image: nginx  
      ports:  
        - containerPort: 80  
---  
apiVersion: v1  
kind: Pod  
metadata:
```

Step 4: Apply Namespace YAML

```
kubectl apply -f namespaces.yaml
```

Step 5: Apply Pod YAML

```
kubectl apply -f pods.yaml
```

```

- ContainerPort: 80
1RV24MC087_SSDhanush@dhhanush-Latitude:~/Program-9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC087_SSDhanush@dhhanush-Latitude:~/Program-9$ kubectl get namespaces
NAME                STATUS    AGE
default             Active   6m21s
dev                 Active   14s
kube-node-lease     Active   6m21s
kube-public          Active   6m21s
kube-system          Active   6m21s
kubernetes-dashboard Active   6m16s
prod                Active   14s
1RV24MC087_SSDhanush@dhhanush-Latitude:~/Program-9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created

```

Step 6: Verify Pods by Namespace

```

kubectl get pods -n dev
kubectl get pods -n prod

```

Step 7: Check Pod Status

```

kubectl get pods --all-namespaces

```

Step 8: Filter Pods Using Labels

Frontend pods in dev namespace

```

kubectl get pods -n dev -l app=frontend

```

Backend pods in prod namespace

```

kubectl get pods -n prod -l app=backend

```

```

frontend-pod 0/1 ContainerCreating 0 31s
1RV24MC087_SSDhanush@dhhanush-Latitude:~/Program-9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           40s
frontend-pod  1/1     Running   0           40s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           40s
frontend-pod  1/1     Running   0           40s
1RV24MC087_SSDhanush@dhhanush-Latitude:~/Program-9$ kubectl get pods --all-namespaces
NAMESPACE     NAME                                READY   STATUS    RESTARTS   AGE
dev            backend-pod                         1/1     Running   0           55s
dev            frontend-pod                        1/1     Running   0           55s
kube-system   coredns-66bc5c9577-k58qb           1/1     Running   0           7m19s
kube-system   coredns-66bc5c9577-s4f2j           1/1     Running   0           7m19s
kube-system   etcd-minikube                       1/1     Running   0           7m24s
kube-system   kube-apiserver-minikube             1/1     Running   0           7m24s
kube-system   kube-controller-manager-minikube    1/1     Running   0           7m25s
kube-system   kube-proxy-k4wrn                    1/1     Running   0           7m19s
kube-system   kube-scheduler-minikube             1/1     Running   0           7m24s
kube-system   storage-provisioner                 1/1     Running   1 (6m48s ago) 7m22s
kubernetes-dashboard dashboard-metrics-scraper-77bf4d6c4c-dp8b4 1/1     Running   0           7m19s
kubernetes-dashboard kubernetes-dashboard-855c9754f9-zbkjk 1/1     Running   0           7m19s
prod          backend-pod                         1/1     Running   0           55s
prod          frontend-pod                        1/1     Running   0           55s

```

Step 9: Cleanup – Delete All Pods

```

kubectl delete pods --all -n dev
kubectl delete pods --all -n prod

```

Step 10: Delete All Services

```
kubectl delete svc --all -n dev
kubectl delete svc --all -n prod
```

```
prod          frontend-pod          1/1          Running          0
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           65s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-9$ kubectl delete pods --all -n dev
kubectl delete pods --all -n prod
pod "backend-pod" deleted
pod "frontend-pod" deleted
pod "backend-pod" deleted
pod "frontend-pod" deleted
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-9$ kubectl delete svc --all -n dev
kubectl delete svc --all -n prod
No resources found
No resources found
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-9$
```